

COMPLETE BEGINNER'S GUIDE

Active Directory Security Lab

with SOC/SIEM Integration

Step-by-Step Setup for Fresh Graduates

	Total Time: 10-12 hours (over a weekend)
	Requirements: VMware Workstation, 16GB RAM, 150GB disk space
	Skills Gained: AD admin, SIEM deployment, threat detection, attack simulation
	Deliverable: Resume-ready GitHub portfolio project

Table of Contents

1. 1. What You Need (Software & Hardware)
2. 2. Phase 1: Download Everything First
3. 3. Phase 2: Configure VMware Networking
4. 4. Phase 3: Build Domain Controller
5. 5. Phase 4: Build Windows 11 Client
6. 6. Phase 5: Install Sysmon (Critical for Detection)
7. 7. Phase 6: Install Splunk (Your SIEM)
8. 8. Phase 7: Create Your First Detection
9. 9. Phase 8: Attack Simulations
10. 10. Phase 9: Documentation & GitHub
11. 11. Troubleshooting Guide
12. 12. Quick Command Reference

1. What You Need

Software Downloads (Get These First)

Software	Where to Get It	Size / Notes
VMware Workstation	You already have this ✓	Installed
Kali Linux	You already have this ✓	Installed
Windows Server 2022	microsoft.com/evalcenter	5.2 GB / 180-day trial
Windows 11 Enterprise	microsoft.com/evalcenter	5.1 GB / 90-day trial
Splunk Enterprise	splunk.com/download	600 MB / Free (8GB/day)
Sysmon	download.sysinternals.com	2 MB / Free Microsoft tool

Hardware Requirements

Component	Requirement
RAM	16GB minimum (20GB recommended) • DC: 4GB Win11: 4GB Splunk: 4GB Kali: 2GB Host: 4GB+
Disk Space	150GB free • DC: 60GB Client: 60GB ISOs: 15GB Splunk: 15GB
CPU	4 cores minimum (8 preferred)

2. Phase 1: Download Everything First

 **Time Required:** 30-60 minutes (depending on internet speed)

Step 1: Create Download Folder Structure

For Linux/Mac Users:

```
mkdir ~/SecurityLab
cd ~/SecurityLab
mkdir ISOs
mkdir Tools
```

For Windows Users:

- Open File Explorer
- Go to your Documents folder
- Create a new folder called 'SecurityLab'
- Inside SecurityLab, create two folders: 'ISOs' and 'Tools'

Step 2: Download All Files

Download these files and place them in the appropriate folders:

Place in ISOs folder:

- ☐ **SERVER_EVAL_x64FRE_en-us.iso** (Windows Server 2022)
 - → <https://www.microsoft.com/en-us/evalcenter/evaluate-windows-server-2022>
- ☐ **Win11_EnterpriseEval.iso** (Windows 11)
 - → <https://www.microsoft.com/en-us/evalcenter/evaluate-windows-11-enterprise>

Place in Tools folder:

- ☐ **splunk-9.x.x-xxxxx-x64-release.msi** (Splunk)
 - → https://www.splunk.com/en_us/download/splunk-enterprise.html
- ☐ **Sysmon.zip**
 - → <https://download.sysinternals.com/files/Sysmon.zip>
- ☐ **sysmonconfig-export.xml**
 - → Right-click Save As from: github.com/SwiftOnSecurity/sysmon-config

3. Phase 2: Configure VMware Networking



Time Required: 10 minutes



WHY THIS MATTERS:

This isolated network prevents your lab attacks from affecting your home network or the internet. It's like creating a sandboxed playground where you can safely practice attacks.

Step 3: Create Custom Virtual Network

In VMware Workstation:

13. Go to Edit → Virtual Network Editor
14. Click 'Change Settings' button (requires administrator privileges)
15. Click 'Add Network' button
16. Select 'VMnet2' (or any unused VMnet number)
17. Configure the following settings:

Type	Host-only
Subnet IP	192.168.100.0
Subnet Mask	255.255.255.0
DHCP	DISABLED (uncheck 'Use local DHCP')

18. Click 'Apply'
19. Click 'OK'



CHECKPOINT: VMnet2 is now created and ready to use!

Understanding the Network Setup



What This Means in Simple Terms:

- **Host-only:** Your VMs can talk to each other and your host computer, but **NOT** the internet
- **192.168.100.x:** This is your lab's private IP address range (like apartment numbers in a building)
- **DHCP Disabled:** You'll manually assign IP addresses (more control, better for learning)

4. Phase 3: Build Domain Controller



Time Required: 2 hours

Step 4: Create Windows Server 2022 VM

In VMware:

20. File → New Virtual Machine
21. Select 'Typical'
22. Click 'Installer disc image file (iso)' → Browse to SERVER_EVAL ISO
23. Guest Operating System:
 - OS: Microsoft Windows
 - Version: Windows Server 2022
24. Name: DC1
25. Disk size: 60 GB (Store as single file)
26. Click 'Customize Hardware':
 - Memory: 4096 MB (4GB)
 - Processors: 2 cores
 - Network Adapter: VMnet2 (the one you created)
 - Remove: Printer, USB Controller (optional)
27. Click 'Finish'

Step 5: Install Windows Server 2022

Start the VM:

28. Power on DC1
29. Press any key to boot from ISO
30. Installation wizard:
 - Language: English
 - Edition: Windows Server 2022 Standard Evaluation (Desktop Experience)
 - Installation Type: Custom
 - Disk: Select the 60GB disk
31. Wait 10-15 minutes for installation
32. When prompted, set Administrator password: P@ssw0rd123!
(Use a complex password you'll remember)
33. Press Ctrl+Alt+Insert to send Ctrl+Alt+Delete
34. Login as Administrator



IMPORTANT:

In VMware, use Ctrl+Alt+Insert instead of Ctrl+Alt+Delete. This is a VMware keyboard shortcut.

Step 6: Configure DC1 Network Settings

On DC1:

35. Right-click Network icon (bottom right) → Open Network & Internet Settings
36. Click 'Change adapter options'
37. Right-click Ethernet0 → Properties
38. Select 'Internet Protocol Version 4 (TCP/IPv4)' → Properties
39. Configure:

IP Address	192.168.100.10
Subnet Mask	255.255.255.0
Default Gateway	(leave blank)
Preferred DNS	192.168.100.10 (itself)
Alternate DNS	(leave blank)

40. Click OK

Step 7: Rename DC1 and Install AD DS

Set Computer Name:

41. Right-click Start → System
42. Click 'Rename this PC'
43. Enter: DC1
44. Click 'Next' → Restart Now

After Reboot - Install Active Directory:

45. Server Manager opens automatically
46. Click 'Add roles and features'
47. Click 'Next' until 'Server Roles'
48. Check 'Active Directory Domain Services'
49. Click 'Add Features' (popup)
50. Click 'Next' → 'Next' → 'Next' → 'Install'
51. Wait 5 minutes for installation

Step 8: Promote to Domain Controller

After installation completes:

52. Click the yellow flag icon (top right of Server Manager)
53. Click 'Promote this server to a domain controller'
54. Deployment Configuration:
 - Select 'Add a new forest'
 - Root domain name: company.local
 - Click 'Next'
55. Domain Controller Options:
 - Forest/Domain functional level: Windows Server 2016
 - Check 'Domain Name System (DNS) server'
 - DSRM Password: P@ssw0rd123!
 - Click 'Next'
56. Click 'Next' through DNS Options
57. NetBIOS name: COMPANY (auto-filled)
58. Click 'Next' → 'Next' → 'Next'
59. Prerequisites check (ignore warnings)
60. Click 'Install'
61. Server will restart automatically (5 minutes)

Step 9: Create Domain User and Security Group

After DC1 reboots:

62. Login as: COMPANY\Administrator (password: P@ssw0rd123!)
63. Open 'Active Directory Users and Computers'
Server Manager → Tools → Active Directory Users and Computers
64. Expand company.local
65. Right-click 'Users' → New → User
66. Create user:
 - First name: Unnathi
 - Last name: Vithlani
 - User logon name: unnathi.vithlani
 - Click 'Next'
 - Password: P@ssw0rd123!
 - Uncheck 'User must change password at next logon'
 - Check 'Password never expires'
 - Click 'Next' → 'Finish'

Create Security Group:

67. Right-click 'Users' → New → Group
68. Group name: SystemAdmins
69. Group scope: Global
70. Group type: Security
71. Click 'OK'

Add User to Group:

72. Right-click 'Unnathi Vithlani' → 'Add to a group'
73. Type: Domain Admins → Click 'Check Names' → OK
74. Right-click 'Unnathi Vithlani' → 'Add to a group'
75. Type: SystemAdmins → Click 'Check Names' → OK

✅ **CHECKPOINT:** DC1 is now a fully functional Domain Controller with DNS!

Test by opening PowerShell and running:

```
Get-ADUser unnathi.vithlani
```

You should see user details displayed!

5. Phase 4: Build Windows 11 Client



Time Required: 1.5 hours

Step 10: Create Windows 11 VM

In VMware:

76. File → New Virtual Machine
77. Select 'Typical'
78. Installer disc image: Browse to Win11_EnterpriseEval.iso
79. Guest OS:
 - OS: Microsoft Windows
 - Version: Windows 10 x64 (Win11 not listed, use Win10)
80. Name: WIN11-CLIENT
81. Disk: 60 GB (single file)
82. Customize Hardware:
 - Memory: 4096 MB
 - Processors: 2 cores
 - Network Adapter: VMnet2 (same as DC)
83. Click 'Finish'

Step 11: Install Windows 11

Start VM:

84. Power on WIN11-CLIENT
85. Press any key to boot
86. Installation:
 - Language: English
 - Click 'Install now'
 - Product key: Skip (click 'I don't have a product key')
 - Edition: Windows 11 Enterprise Evaluation
 - Installation type: Custom
 - Select 60GB disk → Next
87. Wait 10-15 minutes
88. Windows 11 Setup (OOBE):
 - Region: United States
 - Keyboard: US
 - Click 'Skip' for second keyboard
 - Network: Click 'I don't have internet'
 - Click 'Continue with limited setup'
 - Name: LocalUser
 - Password: P@ssw0rd123!
 - Security questions: Answer anything
 - Privacy settings: Turn all OFF → Accept
89. Wait for desktop

Step 12: Configure WIN11-CLIENT Network

On WIN11-CLIENT:

90. Right-click Network icon → Network and Internet settings
91. Click 'Ethernet'
92. Click 'Edit' next to IP assignment
93. Select 'Manual' → Toggle IPv4 ON
94. Configure:

IP Address	192.168.100.20
Subnet prefix length	24
Gateway	(leave blank)
Preferred DNS	192.168.100.10 (the DC)

95. Click 'Save'

Step 13: Test Connectivity

On WIN11-CLIENT, open PowerShell:

```
ping 192.168.100.10
# Should get replies from DC

ping company.local
# Should resolve and get replies

nslookup company.local
# Should show DC IP: 192.168.100.10
```

If ping fails:

On DC1: Run PowerShell as Admin:

```
Set-NetFirewallRule -DisplayName "File and Printer Sharing (Echo Request - ICMPv4-In)" -enabled True
```

Step 14: Join WIN11-CLIENT to Domain

On WIN11-CLIENT:

96. Right-click Start → System
97. Scroll down → Click 'Advanced system settings'
98. Click 'Computer Name' tab
99. Click 'Change'
100. Computer name: WIN11-CLIENT
101. Select 'Domain' radio button
102. Enter: company.local
103. Click 'OK'
104. When prompted for credentials:
 - Username: COMPANY\Administrator
 - Password: P@ssw0rd123!
105. Click 'OK'
106. 'Welcome to company.local domain' → OK
107. 'You must restart' → OK → Close → Restart Now

Step 15: Test Domain Login

After WIN11-CLIENT reboots:

108. Click 'Other user' (bottom left)
109. Login as:
 - Username: COMPANY\unnathi.vithlani
 - Password: P@ssw0rd123!
110. First login takes 2-3 minutes to create profile
111. SUCCESS! You now have a domain-joined client

✅ **CHECKPOINT:** Both VMs running, domain working!

Test by opening Command Prompt on WIN11-CLIENT:

```
whoami
# Should show: company\unnathi.vithlani

echo %LOGONSERVER%
# Should show: \\DC1
```

12. Quick Command Reference

Common PowerShell Commands

Task	Command
Check AD User	Get-ADUser unnathi.vithlani
List All Domain Users	Get-ADUser -Filter *
Check Domain	Get-ADDomain
Test Network	Test-NetConnection 192.168.100.10
View Services	Get-Service Where-Object {\$_.Name -like '*splunk*'}
Check Firewall Rules	Get-NetFirewallRule Where-Object {\$_.Enabled -eq 'True'}

Splunk SPL Queries

Description	SPL Query
View All Events	index=main
Windows Login Events	index=main EventCode=4624
Failed Logins	index=main EventCode=4625
Sysmon Process Creation	index=main EventCode=1
Network Connections	index=main EventCode=3
Filter by Host	index=main host=DC1

Resume-Ready Checklist

- ☐ Lab fully functional (all VMs running)
- ☐ 5+ detection rules working in Splunk
- ☐ Screenshots captured (VMs, Splunk, detections, attacks)
- ☐ GitHub repo created and public
- ☐ README.md written with architecture diagram
- ☐ LinkedIn post about your lab
- ☐ Resume updated with lab details



CONGRATULATIONS!

You now have a professional Active Directory Security Lab with SIEM integration. This is a fantastic portfolio project that demonstrates:

- • Active Directory administration
- • SIEM deployment and configuration
- • Threat detection and analysis
- • Attack simulation and testing
- • Security operations center (SOC) skills
- • Windows event log analysis
- • Network security fundamentals

START TODAY. You can have the basic lab running by tomorrow evening!